Nimit Dave

Boston, MA | +1 (857) 397-2694 | nimitdave3001@gmail.com | LinkedIn | GitHub

SUMMARY

ML engineer with 3+ years delivering production AI solutions across Generative AI, agentic pipelines, and fintech ML. Hands-on across the full model lifecycle: data exploration, feature engineering, training, evaluation, deployment, and drift monitoring. M.S. in Data Analytics Engineering from Northeastern University (GPA: 3.8); experienced in RAG, LLM orchestration, responsible AI guardrails, and AWS-based MLOps.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, Java, C/C++

ML & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, CUDA

LLM & Generative AI: LangChain, LangGraph, vLLM, LoRA/PEFT, RAG, Agentic AI, Multi-Agent Orchestration, Prompt Engineering, Prompt Orchestration, NLTK, spaCy

Retrieval & Ranking: FAISS, Pinecone, Elasticsearch, Semantic Search, Embedding Models, Reranking, BM25, Vector Stores, Retrieval Optimization

MLOps & Tools: MLflow, Apache Airflow, Apache Spark, Docker, Kubernetes, CI/CD, AWS EKS, Snowflake, Databricks

Cloud & Infrastructure: AWS (SageMaker, Bedrock, EC2, GPU Clusters, Lambda), GCP

Evaluation & Observability: Custom Eval Dataset Design, A/B Testing, Drift Detection, Benchmark Suites, Performance Gates, Precision/Recall/F1 Analysis, Model Governance, Responsible AI

Backend & APIs: FastAPI, Flask, REST APIs, Microservices, Feedback Loop Architecture, Streamlit

WORK EXPERIENCE

ML Engineer — August 2025 – Present

Pioneerstek — Boston, MA

- Architected a multi-agent LLM orchestration system using LangGraph and GPT-4o processing 50K+ daily queries, reducing task-resolution time by 35% versus the prior single-agent baseline.
- Fine-tuned a domain-specific instruction model with LoRA/PEFT on a 120K-example proprietary dataset, improving task completion accuracy by 14% and cutting response latency by 40% over raw GPT-4o API calls.
- Designed a hybrid RAG pipeline (dense FAISS retrieval + BM25 reranking) with retrieval optimization; reduced hallucination rate by ~18% on internal benchmarks and lifted user satisfaction from 3.6 to 4.3/5.
- Migrated model serving to AWS EKS with horizontal autoscaling; cut monthly inference costs by ~28% while maintaining 99.9% uptime and adhering to CI/CD and model governance best practices.
- Established team-wide evaluation standards, responsible AI checklists, and staged-rollout gates; mentored 2 junior ML engineers through their first production model deployments.

AI Developer — November 2024 – August 2025

Sirch — Boston, MA

- Benchmarked summarization approaches against a 500+ example human-annotated eval dataset scored on relevance, coherence, and factual accuracy; findings drove final model selection shipped to users.
- Built a real-time routing classifier using content and complexity signals to direct requests between local and hosted models, cutting API costs by ~40% with no measurable drop in eval-set output quality.
- Implemented a semantic-chunking RAG pipeline with prompt orchestration on the OpenAI API; profiled chunk sizes and retrieval depths to achieve sub-2s end-to-end latency at 1,000+ daily users.
- Defined performance gates and production monitoring for all deployed models, establishing regression baselines that caught 3 output-quality regressions before they reached users.

Machine Learning Engineer — May 2024 – November 2024

Blockhouse.app — New York City, NY

- Contributed to end-to-end model lifecycle: validated execution models against 12 months of historical order data, catching 2 regressions before production release and preventing measurable slippage.
- Built a real-time order-routing model leveraging market-state and liquidity features, reducing slippage by 10% and transaction costs by 12% against institutional benchmarks across backtested and live scenarios.
- Designed a weekly feedback loop ingesting live execution outcomes into the training dataset, keeping models current with shifting market conditions and preventing feature drift.
- Rebuilt inference pipeline on AWS GPU clusters with Docker/Kubernetes containerization; fixed a stale-feature timing bug and reduced inference latency by 25%.

PROJECTS

Cerebras Perf Explorer | *React, TypeScript, Vite, SheetJS, Recharts* GitHub

- Built a rollback and safety system for autonomous LLM agents with state checkpointing, human-in-the-loop escalation thresholds, and a resampling mechanism that rerouted off-trajectory behavior to a human operator.
- Lifted agent safety scores from a 20-30% baseline under static monitoring to 70%+ using dynamic resampling, validated against a custom adversarial benchmark suite covering long-horizon edge cases.

evalscope_ext - Universal Benchmark Pruning | *Python, evalscope, PyTorch, CLIP* GitHub

- Built an evalscope extension pruning large LLM/VLM benchmarks to the smallest principled subset that reliably answers go/no-go reliability questions for both sales decisions and engineering regression testing.
- Designed a stratified diversity-sampling strategy with a difficulty prior; validated leave-one-model-out generalization and provably beat random and top-k baselines, plus MMMU image-encoder stress probes.

AI Agent Safety and Control Framework | *Python, LangGraph, Docker* GitHub

- Built an interactive performance-projection dashboard with audience-specific views: a customer go/no-go verdict per model and an engineer latency-vs-throughput inspector with automated anomaly detection.
- Engineered a tolerant column registry and client-side SheetJS parser that ingests sweeps for previously unseen models with zero code edits; shipped as a static React 18 + TypeScript app on Vercel.

EDUCATION

M.S., Data Analytics Engineering May 2024

Northeastern University · Boston, MA · GPA: 3.8 / 4.0

Relevant Coursework: Machine Learning Operations (MLOps), Applied NLP, Statistical Learning, Data Mining, Foundations of Data Analytics

B.E., Mechanical Engineering May 2022

Mumbai University · Mumbai, India · GPA: 3.7 / 4.0

Relevant Coursework: Applied Machine Learning, Artificial Intelligence, Reinforcement Learning, Data Structures & Algorithms